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Self-Supervised Contrastive Learning for Universal ECG Representation with Explainable Arrhythmia Detection

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AGENDA

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Introduction & Background

01

- Cardiovascular disease claims an estimated 17.9 million lives a year — more than any other non-communicable category.
- Supervised deep learning already crosses 97% accuracy on ECG, but every such result assumes a clean digital 1D signal and a fully labelled corpus.
- At an Indian primary health centre the artefact is a thermal-paper printout — scanned, faxed or photographed. Optical digitisation is fragile, and expert annotation is the scarcest resource of all.
- We drop both prerequisites: contrastive self-supervised learning applied directly to the 2D paper image, with Grad-CAM explanations delivered through a lightweight CDSS.

Why It Matters

UN SDG 3 (targets 3.4 & 3.8) calls for diagnostic technology that is affordable and reachable outside large tertiary hospitals — precisely where ECG remains a paper-based workflow.

17.9M

cardiovascular deaths worldwide each year (WHO reporting cycle)

THE CHALLENGE

Problem Statement & Motivation

State-of-the-art ECG models assume a clean digital waveform and a fully labelled training set. Both assumptions fail at the point of care, where the ECG exists only as a noisy paper printout and expert annotation is unaffordable.

01

Brittle optical digitisation

02

Annotation cost blocks deployment

03

Opaque predictions erode clinical trust

WHAT WE SET OUT TO DO

Research Objectives

03

Primary Objective

Learn a universal ECG representation directly from unlabelled 2D paper-ECG images using NT-Xent contrastive pre-training, with no 1D signal extraction anywhere in the pipeline.

Secondary Objective

Quantify label efficiency by fine-tuning the same frozen encoder under 10%, 20%, 40% and 100% labelling budgets on the AAMI five-class task.

Scope & Constraints

In scope: AAMI super-classes N, S, V, F, Q on GMC2024 and MIT-BIH with a ResNet-18 on one T4 GPU. Out of scope: multi-label diagnosis and prospective clinical validation.

Expected Outcome

A deployable CDSS that pairs every prediction with a Grad-CAM heatmap on the original printout, advancing the diagnostic-access goals of UN SDG 3.

PRIOR WORK

Literature Survey

04

Study (Year)	Approach / Method	Key Findings	Limitation / Gap
Rajpurkar (2017); Hannun (2019)	34-layer CNN on single-lead ECG; scaled to 91,232 ambulatory records	Cardiologist-level agreement on rhythm classification	Requires a clean 1D signal and tens of thousands of expert-annotated beats
Qureshi (2020); Mathew (2021)	HSV thresholding and active contours; FCN-UNet trace segmentation	Usable digitisation on lab-quality scans	Fails on glare, faded toner and non-standard grids; the error is non-recoverable downstream
Kiyasseh (2021); Zhai (2021); Mehari (2022)	Contrastive SSL (CLOCS, 1D-CNN) on 1D ECG segments	Roughly 60% lower labelled-data requirement on PTB-XL	Still assumes the 1D signal is already available and clean
Ganeshkumar (2021); Huang (2023)	Grad-CAM and R-peak-guided attention on 12-lead ECG	Class-discriminative explanations for clinicians	Explains digital signals only; silent on optical grid noise

HOW WE APPROACHED IT

Proposed Methodology

05

A two-phase pipeline: contrastive pre-training on unlabelled ECG paper images, then a linear head fine-tuned under restricted label budgets. Implemented in PyTorch 2.0 on a single Google Colab T4 GPU.

1

Data

1,000 unlabelled GMC2024 paper prints; MIT-BIH for cross-database test

2

Augmentation

Resized crop 0.5-1.0, colour jitter, Gaussian blur

3

Pre-Training

ResNet-18 + MLP head, NT-Xent loss, temperature 0.5

4

Fine-Tuning

Frozen encoder, linear head, 10/20/40/100% labels

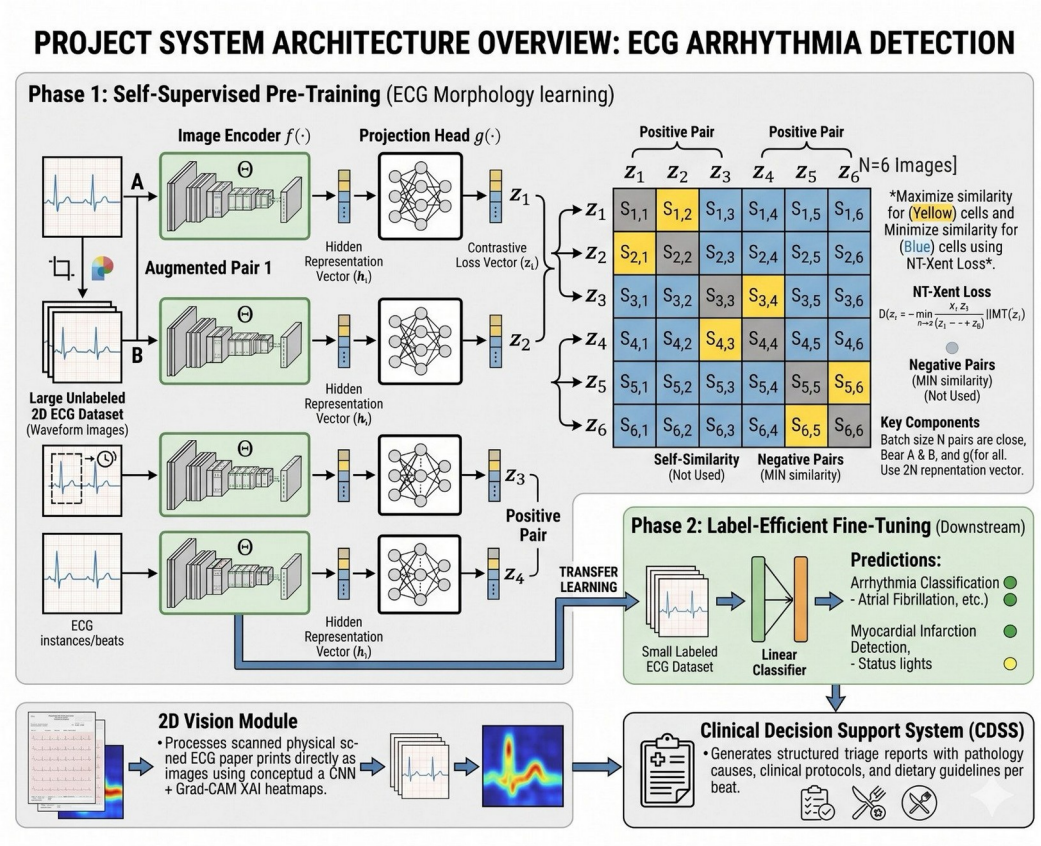
5

XAI & CDSS

Grad-CAM saliency plus an advisory triage report

System Architecture

06



Phase 1 · Contrastive Pre-Training

Two stochastically augmented views of one paper image pass through a shared ResNet-18 f(·) and MLP head g(·). NT-Xent pulls the pair together in a 128-D latent space. No labels are consumed.

Phase 2 · Label-Efficient Fine-Tuning

The projection head is discarded and the 512-D encoder frozen. One linear classifier maps the representation onto the AAMI super-classes N, S, V, F and Q.

2D Vision Module → CDSS

The printout is classified as an image, never reconstructed as a waveform. Grad-CAM localises the driving P, QRS or T region and the CDSS emits an advisory triage report.

Contrastive Pre-Training Convergence


```
🚀 Starting 2D Vision SSL Pre-training on 1000 images...
  Batch 0/15 | Current Loss: 4.9072
🌟 Epoch [1/5] Completed | Average NT-Xent Loss: 4.4298
  Batch 0/15 | Current Loss: 4.2840
🌟 Epoch [2/5] Completed | Average NT-Xent Loss: 4.1293
  Batch 0/15 | Current Loss: 4.0060
🌟 Epoch [3/5] Completed | Average NT-Xent Loss: 3.9587
  Batch 0/15 | Current Loss: 3.7833
🌟 Epoch [4/5] Completed | Average NT-Xent Loss: 3.8681
  Batch 0/15 | Current Loss: 3.7510
🌟 Epoch [5/5] Completed | Average NT-Xent Loss: 3.8975
🎉 Phase 1 Complete: 2D Vision Representations Learned successfully!
```

Phase 1 training console: average NT-Xent loss per epoch on 1,000 unlabelled GMC2024 images.

1,000
unlabelled GMC2024 ECG paper images, no annotation used

4.4298 → 3.8975
average NT-Xent loss, epoch 1 to epoch 5

5 epochs
AdamW, lr 1e-3, batch 64, one Google Colab T4 GPU

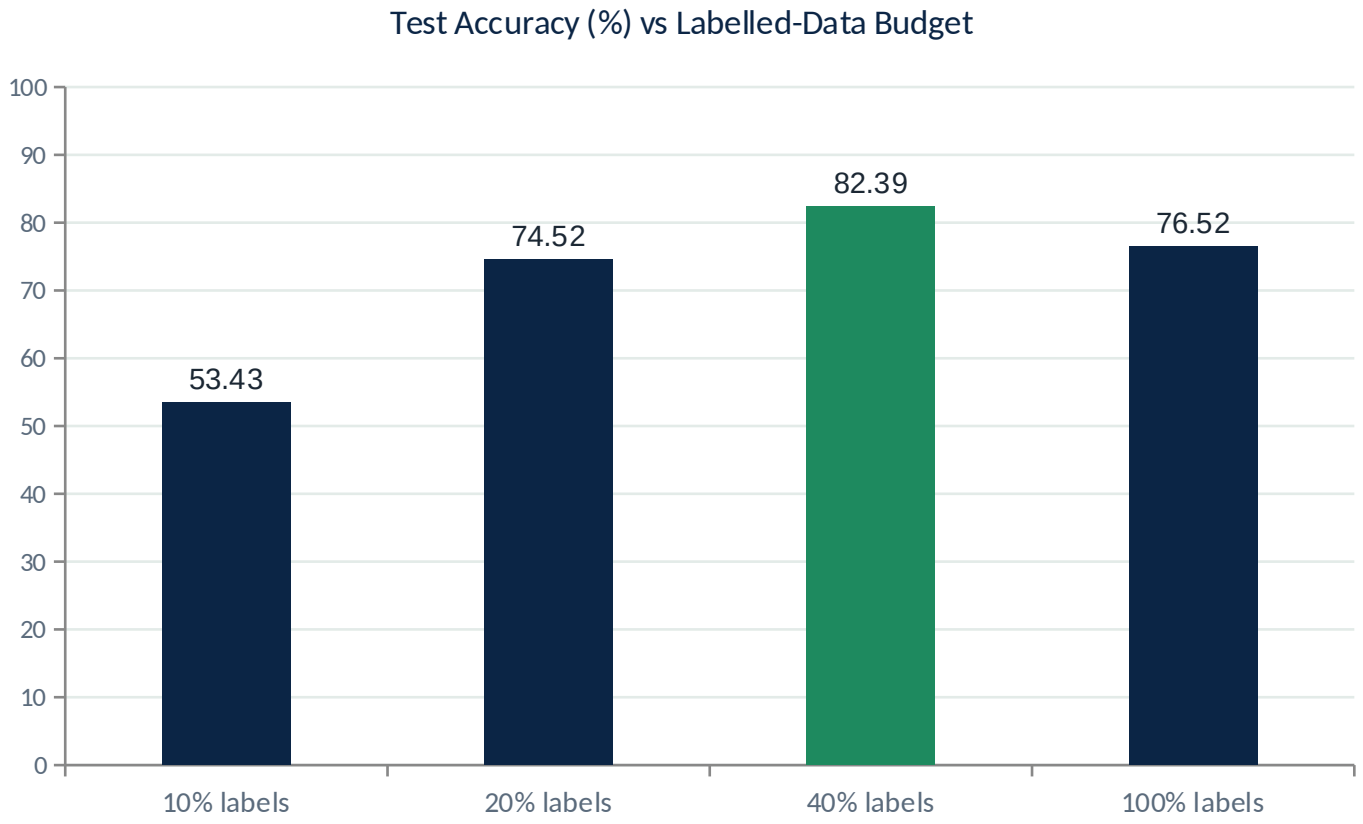
KEY INSIGHT

- The loss falls monotonically across all five epochs — the encoder is separating heartbeat morphology from grid lines and toner artefacts.
- No representation collapse and no loss instability, which we attribute to the strength of the crop–jitter–blur augmentation chain.

WHAT WE FOUND

Results: Label Efficiency

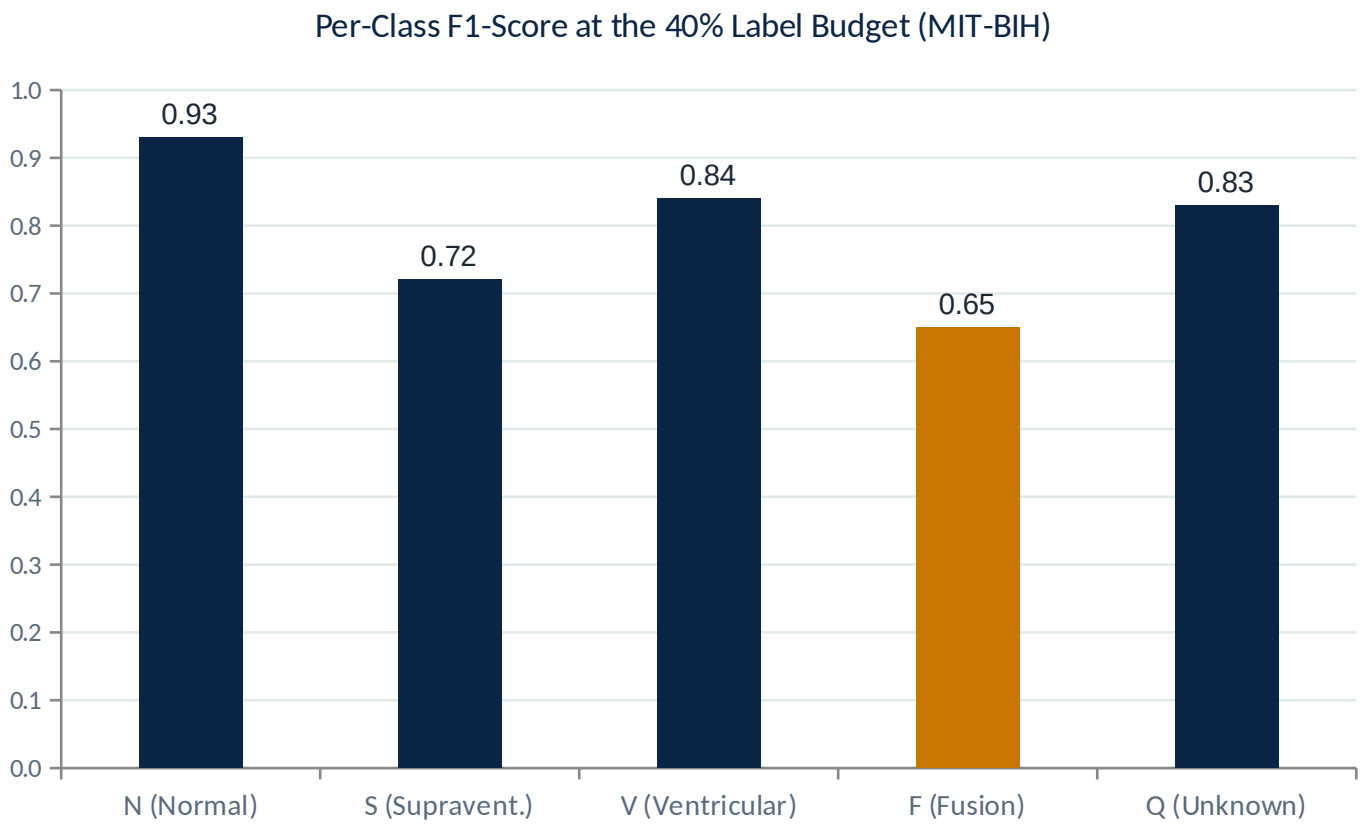
07



KEY FINDINGS

- Peak accuracy of 82.39% with a macro F1 of 0.80 is reached at the 40% labelling budget — using fewer than half the available labels.
- The identical encoder trained on 100% of the labels drops to 76.52%. The SSL features appear to resist the annotation noise concentrated in the ambiguous Fusion and Unknown classes.
- The non-monotonic curve was unexpected, so the experiment was repeated across two further random seeds before it was reported.

Per-Class Performance & Error Analysis



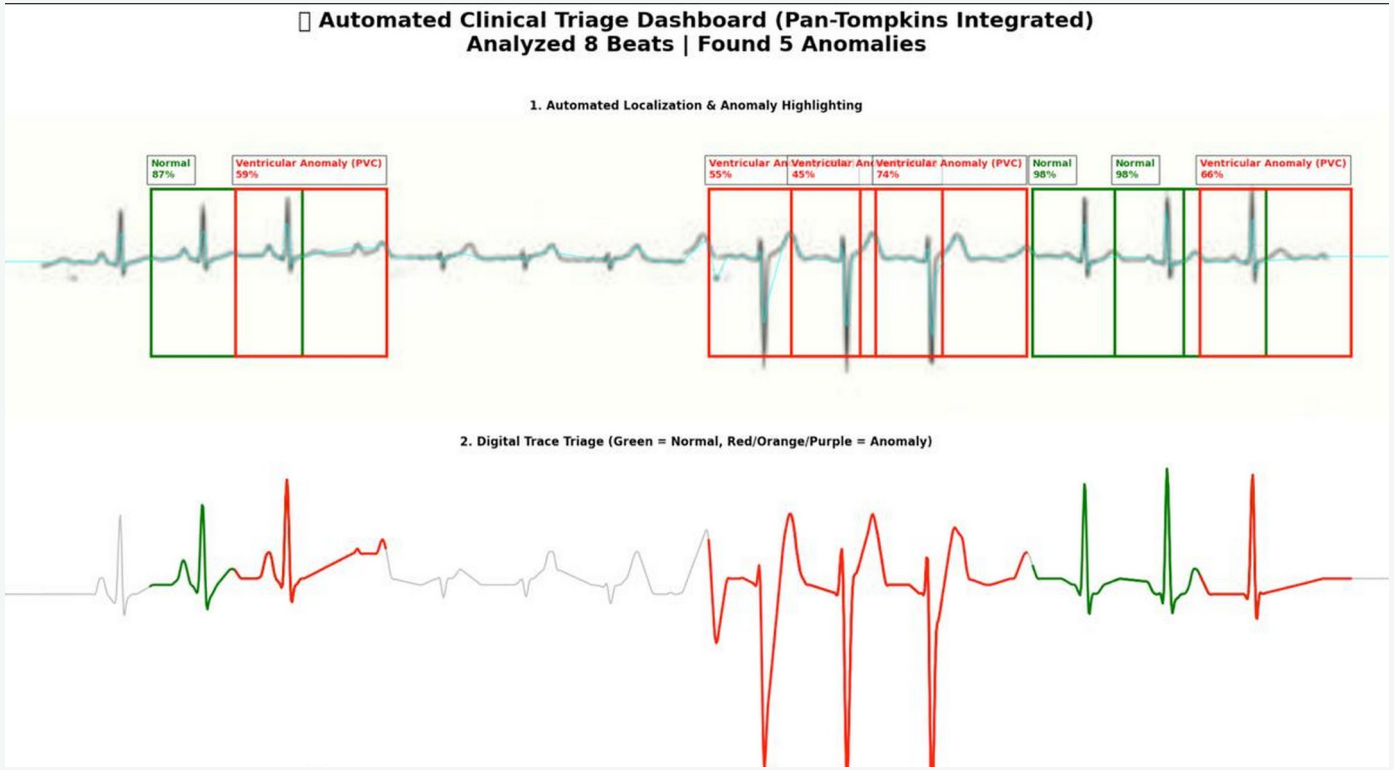
KEY FINDINGS

- At the 40% budget the model records a macro Precision of 0.81, Recall of 0.79 and F1 of 0.80 over 2,409 MIT-BIH test beats (weighted F1 0.88).
- Normal and Ventricular beats resolve cleanly at F1 0.93 and 0.84. Fusion is the weakest class at 0.65, with only 38 test samples and morphology overlapping both N and V.
- The confusion matrix confirms the mechanism: the bulk of fusion errors leak into N and V, the same pattern reported in the clinical literature on fusion-beat ambiguity.

SEEING THE DECISION

Explainable AI — Automated Triage

07



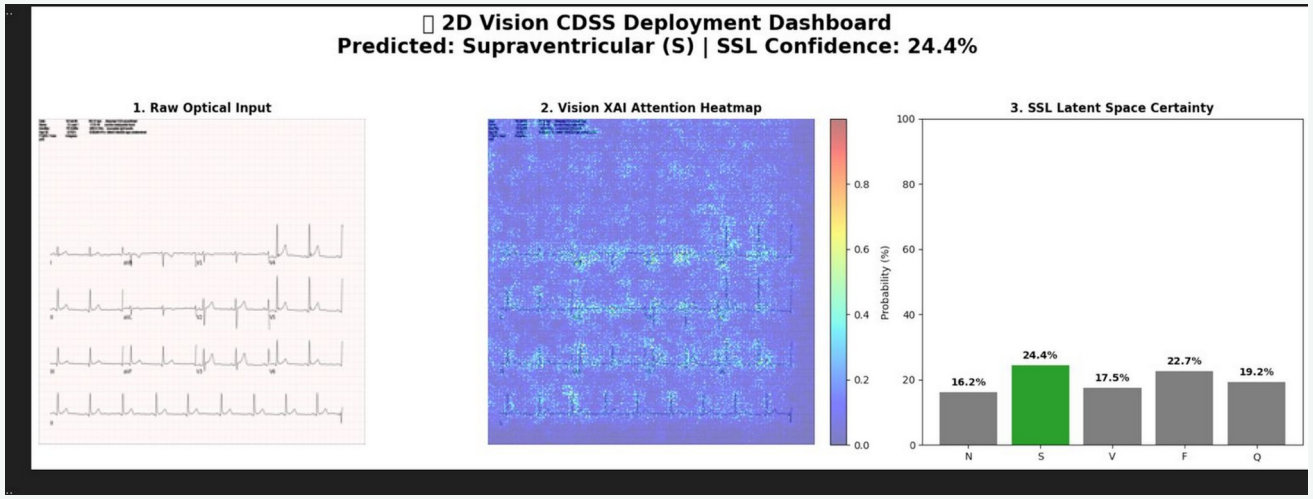
Automated clinical triage dashboard with Pan-Tompkins beat segmentation; green = Normal, red = Ventricular anomaly.

READING THE OUTPUT

- Every beat is localised and boxed, so the clinician scans the strip once and verifies only what was flagged.
- On this 8-beat sample the system flagged 5 Premature Ventricular Contractions, each carrying its own confidence score.
- Per-beat confidence is surfaced rather than hidden, so low-certainty calls are visible instead of silently accepted.
- The report is advisory throughout — it surfaces candidate diagnoses to the cardiologist and never acts as a final classification.

Deployment on a Physical 12-Lead Printout

07



Raw optical input → Grad-CAM attention → per-class SSL latent-space certainty, on a real 12-lead printout.

```
COMPREHENSIVE CLINICAL DECISION SUPPORT REPORT (CDSS)

▶ TOTAL BEATS:        7
▶ FEATURED ANOMALY:   UNKNOWN / PACED BEAT (Q) (Beat 3)
▶ AI CONFIDENCE:      61.80%
▶ ANOMALY ZONE:       Unclassifiable

PATHOLOGY & CAUSES:
   Pacemaker or severe artifact.

CLINICAL PROTOCOL:
   Verify lead placement.

DIETARY & LIFESTYLE GUIDELINES:
   Standard guidelines.
```

The textual CDSS report generated for the same session.

- The Grad-CAM heatmap bypasses the dense grid lines and localises the morphological abnormality — a result no pipeline that must digitise first can reach, because digitisation on this grid fails outright.
- Because saliency is computed on the original image rather than a reconstructed signal, the clinician discusses the explanation on the very artefact they already read.
- The report names the featured anomaly, the confidence, a suspected pathology cause and a recommended next clinical step — advisory in scope, never a final call.

HOW IT STACKS UP

Comparative Analysis

08

Supervised 1D Models	1D Self-Supervised	Ours: 2D SSL on Paper
Input: clean digital 1D signal	Input: clean digital 1D signal	Input: raw paper image, no digitisation
Labels used: 100%	Labels used: 100%	Labels used: 40%
Accuracy: 98.25% / 97.78%	Accuracy: 96.55%	Accuracy: 82.39%, macro F1 0.80
Explanation on signal only	No explanation layer	Grad-CAM plus CDSS on the printout

Key Contributions

Novel Framework

The first contrastive SSL framework applied directly to 2D ECG paper images. The fragile 1D extraction stage is removed from the pipeline entirely.

Label Efficiency

82.39% accuracy with 40% of the labels — roughly six points above the same architecture trained supervised on 100% (76.52%).

Explainable Triage

Grad-CAM heatmaps and a CDSS report on the artefact clinicians already hold, targeting the diagnostic-access goals of UN SDG 3.

Conclusion & Future Scope

Conclusion

- A 2D vision SSL pipeline classifies physical ECG paper images directly, with no 1D extraction stage anywhere in the workflow.
- ResNet-18 pre-trained with NT-Xent on GMC2024 reaches 82.39% accuracy and macro F1 0.80 using 40% of the labels, above the 76.52% fully-supervised baseline.
- Grad-CAM localises the diagnostic feature on the original printout, so the CDSS output is auditable on the same artefact the cardiologist already reads.

Future Scope

- Quantisation-aware training and structured pruning of the ResNet-18 for offline inference on commodity Android devices.
- Federated learning across hospital networks so that patient data never leaves the site of care.
- Broadening beyond the AAMI five classes — MI localisation, bundle-branch blocks, LVH — and incorporating patient context such as age, sex and prior history.

References

[1] Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In Proc. ICML, pp. 1597-1607.

[2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proc. IEEE CVPR, pp. 770-778.

[3] Wagner, P., et al. (2020). PTB-XL, a large publicly available electrocardiography dataset. Scientific Data, 7(1), 154.

[4] Goldberger, A. L., et al. (2000). PhysioBank, PhysioToolkit and PhysioNet. Circulation, 101(23), e215-e220.

[5] Selvaraju, R. R., et al. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In Proc. IEEE ICCV, pp. 618-626.

[6] Mehari, T., & Strodthoff, N. (2022). Self-supervised representation learning from 12-lead ECG data. Computers in Biology and Medicine, 141, 105114.



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